GATE ME 90 Days Challenge

Day 16/90 · Practice Set with Detailed Solutions

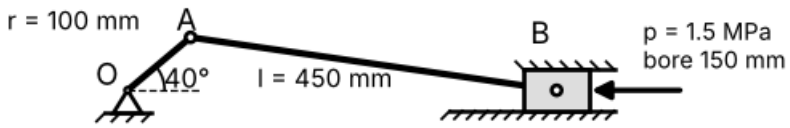
SUBJECT	TODAY'S TOPIC	COUNTDOWN
Theory of Machines	**Dynamics of Linkages & Flywheels**	**93 days to GATE**

How to use this set: 15 questions (7 MCQ, 8 NAT) on static and dynamic force analysis of the slider-crank and four-bar linkages (piston effort, crank effort, inertia force and torque, D'Alembert's principle, equivalent dynamical systems), turning-moment diagrams, fluctuation of energy and speed, and flywheel sizing. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 NAT

In the horizontal engine shown, the crank OA (100 mm) is at 40° from the inner dead centre and the connecting rod AB is 450 mm long. The bore is 150 mm and the net gas pressure on the piston is 1.5 MPa (inertia of the parts neglected). The turning moment on the crankshaft (in kN·m, round off to two decimal places) is _____



Q2 MCQ

For a reciprocating engine with connecting rod 5 times the crank radius, running at uniform speed, the ratio of the inertia force of the reciprocating parts at the inner dead centre to that at the outer dead centre (magnitudes) is

(A) 1.0
(B) 1.22
(C) 1.5
(D) 2.0

Q3 NAT

A vertical engine has its cylinder above the crankshaft. The reciprocating parts weigh 20 kg, the crank radius is 90 mm, the connecting rod is 4 times the crank and the speed is 900 rpm. On the down (expansion) stroke, with the crank at 45° from the top dead centre, the net gas force on the piston is 18 kN downwards. The piston effort (in kN, round off to two decimal places) is _____

Q4 NAT

An engine runs at 3000 rpm. Its reciprocating parts have a mass of 1.5 kg, the crank radius is 60 mm and the connecting rod is 3.5 times the crank. The inertia torque on the crankshaft due to the reciprocating parts when the crank is at 45° from the inner dead centre (in N·m, round off to one decimal place) is _____

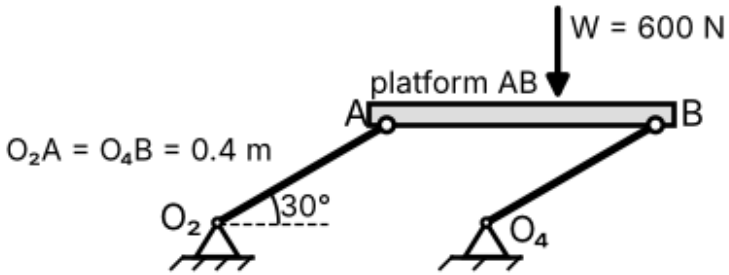
Q5 MCQ

A mechanical press is driven by a crank of radius 50 mm and a connecting rod of 200 mm. The ram (slider) must exert 50 kN when the crank is 10° before the dead centre at which the ram reaches the end of its working stroke. Neglecting friction and inertia, the torque needed on the crank is nearly

(A) 434 N·m
(B) 541 N·m
(C) 2500 N·m
(D) 2443 N·m

Q6 MCQ

The platform AB of the parallel linkage shown carries a load W = 600 N. Links O_2A and O_4B are each 0.4 m long and make 30° with the horizontal; only O_2A is driven. Neglecting the weights of the links and friction, the torque on O_2A needed to hold the platform is



(A) 120.0 N·m
(B) 240.0 N·m
(C) 103.9 N·m
(D) 207.8 N·m

Q7 NAT

To find its radius of gyration, a connecting rod is suspended from a knife edge through the small-end centre and allowed to swing as a compound pendulum. Its centre of mass is 160 mm below the knife edge and the measured period of small oscillations is 0.9 s. Taking g = 9.81 m/s², the radius of gyration of the rod about its centre of mass (in mm, round off to one decimal place) is _____

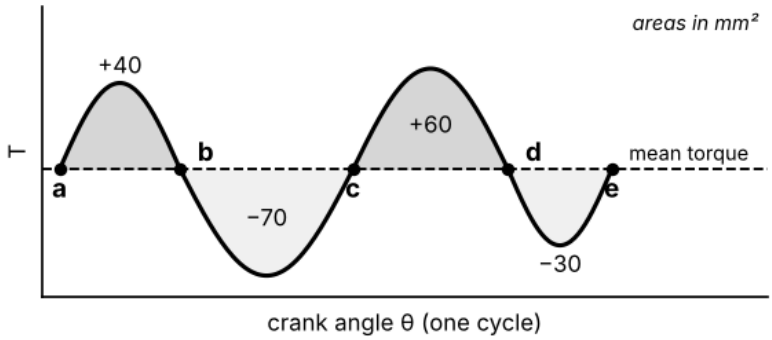
Q8 MCQ

A flywheel of mass 250 kg and radius of gyration 0.5 m runs between 196 rpm and 204 rpm in each cycle. The maximum fluctuation of energy is nearly

(A) 1096.6 J
(B) 548.3 J
(C) 2193.2 J
(D) 274.2 J

Q9 MCQ

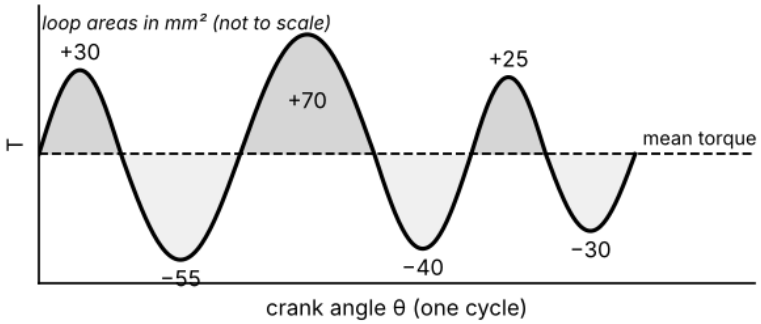
The turning-moment diagram of an engine for one cycle is shown, with the loop areas (in mm²) between the torque curve and the mean-torque line marked: +40 (a to b), −70 (b to c), +60 (c to d) and −30 (d to e). The flywheel speed is maximum and minimum, respectively, at



(A) d and c
(B) b and c
(C) b and e
(D) d and a

Q10 NAT

The turning-moment diagram of an engine for one cycle is shown. The loop areas above and below the mean-torque line, in order, are +30, −55, +70, −40, +25 and −30 mm². The scales are 1 mm = 100 N·m and 1 mm = 5°. The engine runs at a mean speed of 150 rpm and the speed must stay within ±2 % of the mean. The moment of inertia of the flywheel required (in kg·m², round off to two decimal places) is _____



Q11 NAT

The turning moment of a multi-cylinder engine is T = 2000 + 800 sin 3θ N·m, where θ is the crank angle, and the resisting torque is constant. If the total fluctuation of speed must not exceed ±1 % of the mean speed of 300 rpm, the moment of inertia of the flywheel required (in kg·m², round off to two decimal places) is _____

Q12 NAT

A four-stroke engine develops 20 kW at 600 rpm. The coefficient of fluctuation of energy is 0.1 and the speed must be kept within ±1 % of the mean speed. The moment of inertia of the flywheel required (in kg·m², round off to two decimal places) is _____

Q13 MCQ

In a six-cylinder four-stroke engine with evenly spaced firing, the number of main fluctuation cycles of the turning-moment diagram per revolution of the crankshaft is

(A) 6
(B) 1.5
(C) 12
(D) 3

Q14 NAT

A machine has a maximum fluctuation of energy of 12 kJ. Its flywheel runs at a mean speed of 200 rpm and the speed must stay within ±1.5 % of the mean. The rim, of mean radius 0.75 m, is to provide 90 % of the required moment of inertia (the hub and arms provide the rest). The mass of the rim (in kg, round off to the nearest whole number) is _____

Q15 MCQ

A cast-iron rim flywheel (density 7200 kg/m³) may have a maximum hoop stress of 6 MPa due to rotation. Treating it as a thin rim, the largest mean diameter it can have if it runs at 300 rpm is nearly

(A) 0.92 m
(B) 3.68 m
(C) 1.84 m
(D) 1.20 m

Answer Key

Q	Type	Answer
Q1	NAT	2.00 (accept 1.98 to 2.02)
Q2	MCQ	(C) 1.5
Q3	NAT	6.89 (accept 6.85 to 6.93)
Q4	NAT	320.3 (accept 318 to 322.5)
Q5	MCQ	(B) 541 N·m
Q6	MCQ	(D) 207.8 N·m
Q7	NAT	81.3 (accept 80.8 to 81.7)
Q8	MCQ	(A) 1096.6 J
Q9	MCQ	(B) b and c
Q10	NAT	61.89 (accept 61.5 to 62.3)
Q11	NAT	27.02 (accept 26.85 to 27.2)
Q12	NAT	5.07 (accept 5.03 to 5.10)
Q13	MCQ	(D) 3
Q14	NAT	1459 (accept 1450 to 1468)
Q15	MCQ	(C) 1.84 m

Detailed Solutions

Q1 **Answer: 2.00 (accept 1.98 to 2.02)**

Concept: Static force analysis of the slider-crank: piston effort F_P → rod thrust $F_Q = F_P/\cos\beta$ → crank effort $F_T = F_Q \sin(\theta + \beta)$ → torque $T = F_T\, r$.

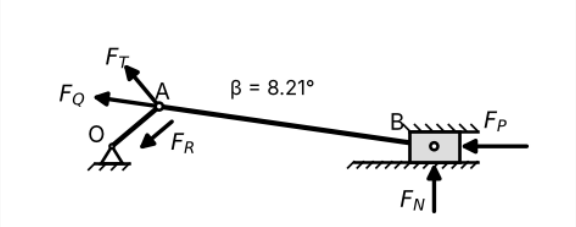
$F_P = p \times \pi d^2/4 = 1.5 \times \pi \times 150^2/4 =$ **26 507 N**

$n = l/r = 4.5$; $\sin\beta = \sin 40°/4.5 = 0.1428 \rightarrow$ **$\beta = 8.21°$**

$F_Q = 26\,507/\cos 8.21° =$ **26 782 N**; side thrust $F_N = F_P \tan\beta = 3826$ N

$F_T = 26\,782 \times \sin 48.21° =$ **19 969 N**; $T = 19\,969 \times 0.1 = 1997$ N·m ≈ **2.00 kN·m**

Exam tip: The force diagram (figure) shows how the rod thrust at A splits into F_T (turns the crank) and F_R (loads the main bearing). Shortcut: $T = F_P\, r \sin(\theta + \beta)/\cos\beta$.



Q2 **Answer: (C) 1.5**

Concept: $F_I = m\,r\,\omega^2 (\cos\theta + \cos 2\theta/n)$.

Inner dead centre, $\theta = 0°$: $F_I = m\,r\,\omega^2 (1 + 1/5) =$ **$1.2\ m\,r\,\omega^2$**

Outer dead centre, $\theta = 180°$: $F_I = m\,r\,\omega^2 (-1 + 1/5) \rightarrow$ magnitude **$0.8\ m\,r\,\omega^2$**

Ratio = 1.2/0.8 = **1.5**

Exam tip: With an infinitely long rod (pure SHM) the ratio would be 1. Obliquity makes the piston accelerate harder at the dead centre nearer the cylinder head.

Q3 **Answer: 6.89 (accept 6.85 to 6.93)**

Concept: Piston effort F_P = gas force − inertia force ± weight. On the down stroke the weight of the reciprocating parts helps the gas force.

$\omega = 2\pi \times 900/60 =$ **94.25 rad/s**; $m\,r\,\omega^2 = 20 \times 0.09 \times 94.25^2 =$ **15 989 N**

$F_I = 15\,989 \times (\cos 45° + \cos 90°/4) = 15\,989 \times 0.7071 =$ **11 306 N** (opposes the motion, since the piston is still accelerating)

$F_P = 18\,000 - 11\,306 + 20 \times 9.81 =$ **6890 N = 6.89 kN**

Exam tip: For the up stroke of a vertical engine, the weight is subtracted instead. Before about 75° to 80° of crank angle the inertia force reduces the piston effort; near the end of the stroke it adds to it.

Q4 **Answer: 320.3 (accept 318 to 322.5)**

Concept: Inertia torque $T_I = F_I \times r\,(\sin\theta + \sin 2\theta/(2n))$, the same lever arm as for the piston effort (from virtual work, v_{piston}/ω).

$\omega = 314.16$ rad/s; $m\,r\,\omega^2 = 1.5 \times 0.06 \times 314.16^2 =$ **8883 N**

$F_I = 8883 \times (\cos 45° + \cos 90°/3.5) =$ **6281 N**

Lever arm $= 0.06 \times (\sin 45° + \sin 90°/7) = 0.06 \times 0.8500 =$ **0.0510 m**

$T_I = 6281 \times 0.0510 =$ **320.3 N·m**

Exam tip: The inertia torque averages to zero over a revolution: it does no net work, but it adds to the fluctuation of the turning moment at high speeds.

Q5 **Answer: (B) 541 N·m**

Concept: $T = F\,r \sin(\theta + \beta)/\cos\beta$ with θ measured from the dead centre; it tends to zero as the crank approaches the dead centre (toggle action).

$n = 200/50 = 4$; $\sin\beta = \sin 10°/4 \rightarrow$ **$\beta = 2.49°$**

$T = 50\,000 \times 0.05 \times \sin 12.49°/\cos 2.49° = 2500 \times 0.2163/0.9991 =$ **541 N·m**

Exam tip: 434 N·m ignores the rod obliquity and 2500 N·m is F r (the value at 90°). Near dead centre a small torque gives a large ram force, which is why presses and toggle clamps work there.

Q6 **Answer: (D) 207.8 N·m**

Concept: Virtual work: $T\,\delta\theta = W\,\delta y$. In a parallelogram linkage the platform translates, so every point of it moves like A.

Point A moves on a circle of radius r; its vertical velocity = $\omega\,r\cos 30°$.

$T\,\omega = W\,\omega\,r\cos 30° \rightarrow T = 600 \times 0.4 \times 0.866 =$ **207.8 N·m**

Exam tip: The position of W on the platform does not matter, because the platform translates. 120 N·m uses sin 30° instead of cos 30°.

Q7 **Answer: 81.3 (accept 80.8 to 81.7)**

Concept: Compound pendulum: $T_p = 2\pi\sqrt{[(k_G^2 + h^2)/(g\,h)]}$, so the equivalent simple-pendulum length is $L = (k_G^2 + h^2)/h$.

$L = g\,T_p^2/(4\pi^2) = 9.81 \times 0.81/39.48 =$ **0.2013 m**

$k_G^2 = L\,h - h^2 = 0.2013 \times 0.16 - 0.0256 =$ **0.006 608 m²**

$k_G =$ **0.0813 m = 81.3 mm**

Exam tip: k_G is needed for the dynamically equivalent two-mass model of the rod ($l_1\,l_2 = k_G^2$). Do not forget to subtract h^2.

Q8 **Answer: (A) 1096.6 J**

Concept: $\Delta E = \tfrac{1}{2} I\,(\omega_{max}^2 - \omega_{min}^2) = I\,\omega_{mean}\,(\omega_{max} - \omega_{min})$.

$I = m\,k^2 = 250 \times 0.5^2 =$ **62.5 kg·m²**

$\omega_{mean} = 2\pi \times 200/60 = 20.944$ rad/s; $\omega_{max} - \omega_{min} = 2\pi \times 8/60 = 0.8378$ rad/s

$\Delta E = 62.5 \times 20.944 \times 0.8378 =$ **1096.6 J**

Exam tip: Equivalently $\Delta E = I\,\omega^2\,C_s$ with $C_s = 8/200 = 0.04$. 2193.2 J drops the ½, and 548.3 J uses half the speed range.

Q9 **Answer: (B) b and c**

Concept: The flywheel speeds up while $T > T_{mean}$ and slows down while $T < T_{mean}$. Track the energy relative to the start point a.

Energy at the crossings: b = E + 40, c = E − 30, d = E + 30, e = E + 0.

Maximum energy (speed) at **b**, minimum at **c**; $\Delta E = 40 - (-30) = 70$ mm².

Exam tip: The largest loop (+60) does not decide the maximum: what counts is the running total. Speed extremes always lie at crossings of the mean-torque line.

Q10 **Answer: 61.89 (accept 61.5 to 62.3)**

Concept: ΔE = (maximum − minimum of the cumulative energy) × area scale; then $I = \Delta E/(\omega^2 C_s)$.

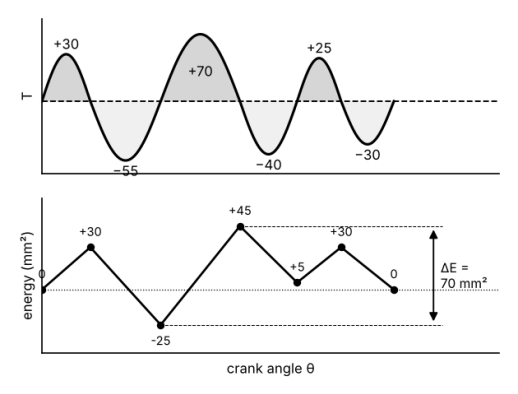
Running total of areas: 0, +30, −25, +45, +5, +30, 0 → max +45, min −25 → **ΔE = 70 mm²** (see the energy curve).

1 mm² = 100 × (5 × π/180) = **8.727 J**, so ΔE = 70 × 8.727 = **610.9 J**

$\omega = 2\pi \times 150/60 = 15.708$ rad/s; $C_s = 0.04$

$I = 610.9/(15.708^2 \times 0.04)$ = **61.89 kg·m²**

Exam tip: ±2 % means $C_s = 0.04$, not 0.02. Convert the angle scale to radians before multiplying the scales.



Q11 **Answer: 27.02 (accept 26.85 to 27.2)**

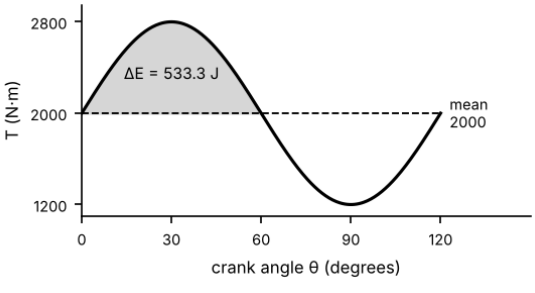
Concept: With a constant resisting torque, $T_{resisting} = T_{mean}$ = 2000 N·m. The excess loop is where 800 sin 3θ > 0, i.e. 0 ≤ θ ≤ 60°.

$\Delta E = \int_0^{\pi/3} 800 \sin 3\theta \, d\theta = 800 \times 2/3$ = **533.3 J** (shaded area)

$\omega = 2\pi \times 300/60 = 31.416$ rad/s; $C_s = 0.02$

$I = 533.3/(31.416^2 \times 0.02)$ = **27.02 kg·m²**

Exam tip: For T = A + B sin kθ the excess energy is always 2B/k. The mean term A sets the power (P = A ω = 62.8 kW here), not the flywheel size.



Q12 **Answer: 5.07 (accept 5.03 to 5.10)**

Concept: C_e = ΔE / (work done per cycle); a four-stroke cycle takes two revolutions.

Cycles per minute = 600/2 = 300 → work per cycle = 20 000 × 60/300 = **4000 J**

ΔE = 0.1 × 4000 = **400 J**

$I = \Delta E/(\omega^2 C_s) = 400/(62.83^2 \times 0.02)$ = **5.07 kg·m²**

Exam tip: Using 600 cycles/min (two-stroke thinking) halves ΔE and gives 2.53 kg·m². Read whether the engine is two-stroke or four-stroke.

Q13 **Answer: (D) 3**

Concept: Each cylinder fires once every 720° of crank rotation in a four-stroke engine.

Six cylinders → firing interval = 720°/6 = **120°**

Number of power impulses (fluctuation cycles) per 360° = 360/120 = **3**

Exam tip: More, closer impulses mean smaller excess-energy loops and a lighter flywheel, which is why multi-cylinder engines need small flywheels. A six-cylinder two-stroke engine would give 6.

Q14 **Answer: 1459 (accept 1450 to 1468)**

Concept: Total $I = \Delta E/(\omega^2 C_s)$; a thin rim has $I_{rim} = m R^2$.

ω = 2π × 200/60 = 20.944 rad/s; C_s = 0.03

$I = 12\,000/(20.944^2 \times 0.03)$ = **911.9 kg·m²**

I_{rim} = 0.9 × 911.9 = 820.7 kg·m² → $m = 820.7/0.75^2$ = **1459 kg**

Exam tip: The rim does almost all the work because its mass is far from the axis. Check the rim speed too: v = ωR = 15.7 m/s here, safe for cast iron.

Q15 **Answer: (C) 1.84 m**

Concept: In a thin rotating rim the hoop stress is $\sigma = \rho v^2$, where v = ω R is the rim speed (it does not depend on the cross-section).

$v_{max} = \sqrt{(\sigma/\rho)} = \sqrt{(6 \times 10^6/7200)}$ = **28.87 m/s**

ω = 31.416 rad/s → R = 28.87/31.416 = 0.919 m → D = **1.84 m**

Exam tip: 0.92 m is the radius, not the diameter. Since the stress limits the rim speed, a flywheel cannot be made more effective without limit just by making it larger.

What next?

Day 16 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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